Departure and destination aerodromes in the OPDI flight list

How they are derived, how accurate they are, and how to use them

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Who this is for

You are reading this because you use, or are deciding whether to use, the `ADEP` and `ADES` columns of the OPDI flight list. It covers what those columns mean, how the pipeline derives them from raw ADS-B, how often they are right, where they are systematically wrong, and what a downstream analysis should do about it.
Every figure on this page is produced by running the pipeline over a sample of ADS-B and scoring the result against EUROCONTROL Network Manager movement data. Nothing is estimated or carried over from an earlier version.

Summary

OPDI reconstructs flights from ADS-B state vectors alone: no flight plans, no schedules, no radar. From the resulting trajectory it must infer where each flight departed from and where it arrived — which is straightforward when the aircraft is seen on the runway at both ends, and hard in every other case.

Three findings shape what the flight list now contains.

Departures and arrivals want different algorithms. A departing aircraft's first ADS-B fix is usually on or near the runway, so geometry alone settles it. An arriving aircraft's last fix is often where reception was lost rather than where it landed, so geometry alone does not. The flight list therefore derives departures with one method and arrivals with another.

Candidate aerodromes are chosen by exact distance. The published algorithm narrowed its candidates using a hexagonal spatial index and measured exact distance only among the survivors, so an aerodrome slightly further out was discarded before its distance was ever computed. Measuring first and choosing afterwards is the single change that makes everything else in this study work.

A departure's null is now two different answers. `ADEP` can be an aerodrome, the marker `OOA` for a flight that began outside the observed area, or null for "could not tell", and a companion column says which. Earlier releases collapsed the last two into a single null, and about one flight in twelve genuinely belongs to the middle one.

Arrivals do not carry that marker, and the asymmetry is deliberate. The label is decided by whether a trajectory's endpoint sits at the edge of coverage. A track that *starts* at the edge almost certainly entered from outside; a track that *ends* near it may equally be a flight still bound somewhere inside whose reception was simply lost. Measured, the arrival label was right about half the time, so it is not emitted. Section 10 gives the figures and Section 9 what would change it.

1 Reading this page

Section 2 says what the flight list contains. Section 3 documents the pipeline that produces it, step by step. Section 4 defines the measures. Everything after that is evidence.

Terms are defined where they first appear and not before. Two conventions run throughout:

- **legacy** means the algorithm and constants behind every OPDI flight list published before 2026 — reachable in code as `DetectionConfig.legacy()`, so released data stays reproducible.
- **shipped** means what this study recommends and what the pipeline now produces by default.

"Production" is used only to talk about the running system, never as the name of an algorithm.

2 What the flight list contains

One row per flight, where a *flight* is a continuous stretch of ADS-B reception from one aircraft under one callsign. The columns this page is about:

Column	Meaning
`ID`	The track identifier. Stable within a release, **not** stable across algorithm versions — see Section 16.
`ICAO24`	The aircraft's 24-bit transponder address. The join key to most other aviation data.
`FLT_ID`	The broadcast callsign. Space-padded to eight characters in the raw feed; trim before joining.
`DOF`	Date of flight, taken from the first reception.
`ADEP` / `ADES`	Departure and destination aerodrome, ICAO code — or `OOA`, or null.
`ADEP_SOURCE` / `ADES_SOURCE`	`aerodrome`, `out_of_area`, or `undetermined`.
`ADEP_P` / `ADES_P`	Other aerodromes that were plausible. Comma-separated, ranked behind the chosen one.
`FIRST_SEEN` / `LAST_SEEN`	Reception bounds. Not off-block and in-block times.
`VERSION`	Which algorithm produced the row.

2.1 The three states of an aerodrome column

This distinction matters more than anything else on this page for a downstream analysis, and it is new:

`aerodrome` An ICAO code. The pipeline saw enough to name it.

`out_of_area` `ADEP` is `OOA`. The flight entered the observed area rather than departing inside it — the trajectory begins at the edge of coverage, not near any aerodrome. **This is an answer, not a failure.** It appears on departures only: see the summary, and Section 10 for why. An analysis counting movements should exclude these; one measuring coverage should count them as correct.

`undetermined` `ADEP`/`ADES` is null. The pipeline could not tell. Filter these out; do not impute them.

Collapsing the last two — as every release before `v4.0.0` did — makes traffic appear to be missing where in fact it was correctly identified as originating elsewhere.

3 The pipeline that produces it

Everything below is real ADS-B, processed by the code in the `opdi` repository. No step is a re-implementation for the purposes of this report.

The same pipeline, step by step. Each row names the module and the table it writes; every step reads the table written above it unless stated otherwise.

Step	Module	Writes	Does
—	—	*OpenSky archive*	1 Hz global state vectors
00	`reference/`	`h3_airport_detection_zones`	H3 aerodrome zones, OurAirports reference
01	`ingestion/osn_statevectors.py`	`osn_statevectors_v2`	European bounding box, thinning to 5 s
02	`pipeline/tracks.py`	`osn_tracks`	Split into tracks, attach the H3 index
02a	`cleaning/cleaner.py`	`osn_tracks_clean`	Mask implausible values, keep the rows
03	`pipeline/flights.py`	`opdi_flight_list`	**ADEP / ADES detection** — reads 02a and 00
04	`pipeline/events.py`	events, measurements	Milestones along each flight
05	`output/`	published dataset	Parquet and CSV export

3.1 00 — Reference data

Builds what the detection needs to know about aerodromes. The important product is the **H3 detection zone table**: for every aerodrome, the set of hexagonal cells (resolution 7, about 1.5 NM across) covering concentric rings out to a generous radius, each tagged with the ring's inner and outer distance.

This is a spatial index, not an answer. Its job is to turn "which aerodromes might this fix be near?" into a hash lookup. Distances are measured afterwards, exactly — Section 3.5.1 explains why that distinction turned out to matter more than any threshold in the study.

Aerodrome positions, elevations and scheduled-service flags come from OurAirports. The candidate set is restricted to large and medium aerodromes; Section 17 quantifies what that excludes.

3.2 01 — State-vector ingestion

Reads the OpenSky archive and keeps what OPDI needs:

- **A bounding box**, roughly 25.9° W to 49.7° E and 26.7° N to 70.3° N. This is the *observed area*, and its edge is what the `OOA` marker refers to.
- **Thinning to 5 s.** The archive is a complete 1 Hz grid. OPDI keeps the newest row in each 5 s bin — a **bin-based** rule. Earlier releases used a fixed-phase rule, keeping a row only if one existed at the exact second divisible by five. Section 12 measures the difference.

The raw global 1 Hz feed is never stored.

3.3 02 — Track building

Groups state vectors into flights and assigns `track_id`. Two aircraft transmissions belong to the same track when they share an aircraft address and callsign and are not separated by a gap of more than 30 minutes, or more than 15 minutes below 5,000 ft — the second catching an aircraft that keeps its transponder on while parked.

This rule is frozen. Changing it changes `track_id` for all published data, so it is not tuned and not swept.

The step also attaches each fix's H3 cell at resolution 7, which is what makes the aerodrome join in step 03 a lookup rather than a scan.

3.4 02a — Trajectory cleaning

Masks implausible values to null and keeps the row: duplicate state vectors, out-of-range positions, stale broadcasts (ADS-B sends position and velocity in separate messages, so identical consecutive values mean *repeated*, not *measured*), derivative spikes, and isolated points. It also marks segment boundaries at coverage gaps so nothing downstream interpolates across them.

Masking rather than deleting is deliberate: it preserves the distinction between "no data" and "interpolated data". It does mean the detection has fewer usable samples, since a fix with no barometric altitude cannot vote on a trend. Section 11 measures what that costs and what it buys.

3.5 03 — Flight list

Where `ADEP` and `ADES` are decided. Two algorithms, one per role.

3.5.1 What both share

Candidate selection. Every aerodrome whose detection zone the aircraft passed through is a candidate. The exact great-circle distance from the trajectory's closest approach to each candidate is computed, and the smallest wins. **No candidate is eliminated before its distance is measured.**

That sentence is the study's central finding, and it was not true before. The legacy algorithm kept only the candidates in the innermost occupied hexagon ring, then measured exact distance among those. A ring steps about 2.8 NM, so an aerodrome a kilometre further out was routinely discarded unmeasured — and the higher the aircraft, the more aerodromes share a ring and the less a ring index can separate them. Section 5 shows what that was worth.

The scheduled-service penalty. An aerodrome without scheduled commercial service has a fixed distance added to its effective distance before that comparison, so a military or general-aviation field must be *clearly* nearest rather than merely nearest. It re-ranks candidates without changing how many flights are answered, so it can move accuracy and cannot move coverage.

3.5.2 `endpoint` — for departures

Take the aerodrome nearest the track's first fix, and accept it only if that fix is within a radius and either on the ground or below a height *above field elevation* — above the aerodrome's own elevation, since a fixed altitude cut-off means nothing at a field sitting at 5,000 ft. Failing that, if the fix sits at the edge of the observed area, answer `OOA`; otherwise abstain.

Its abstention is **geometric**: it declines when the endpoint is not where a departure's endpoint should be.

3.5.3 `trend` — for arrivals

For every fix below a flight-level cap and within a radius of a candidate aerodrome, ask whether smoothed barometric altitude is rising or falling. Each fix votes. If descending votes beat climbing votes by more than a margin, that aerodrome is a destination.

Its abstention is **evidential**: it declines when the altitude trace near an aerodrome does not clearly rise or fall. That is why it suits arrivals — it uses the whole approach rather than one fix, and one fix is exactly what an arrival is least likely to have in the right place.

Two refinements complete it. Among candidates whose distances differ by less than a small band, the one the aircraft's course actually points at wins (Section 13). And where `trend` cannot name an aerodrome, the same border test `endpoint` uses can still distinguish "came from outside" from "cannot tell".

3.6 04–05 — Events and export

The flight list becomes the spine for milestone extraction — take-off, landing, top of climb, level segments, flight-level crossings — and is exported as parquet and CSV. Neither step changes `ADEP` or `ADES`.

4 How the numbers are measured

Two stages, and the distinction governs how much any figure is worth.

The research sweep. Hundreds of parameter combinations applied to a cached intermediate table. Cheap enough to be exhaustive, which is what makes a grid of this size possible — but it re-implements the decision rule rather than running it, so it can only *propose* a value.

The pipeline run. The proposed values run through the same code path that writes the published flight list, and scored identically. Nothing in this report becomes a default without passing this stage.

4.1 The sample

Three days, **5–7 June 2025**, with **5–7 June 2024** as an independent second period. Both are full-day samples inside the bounding box, thinned to 5 s.

Ground truth is EUROCONTROL Network Manager movement data for the same days, matched to OPDI tracks on aircraft address plus callsign plus date, and where several tracks match, the one starting closest to the reference off-block time. It is not redistributable, which is where reproducibility stops (Section 18).

4.2 The measures

For each role — departures and arrivals — separately:

Coverage Of the reference flights, the share OPDI gives *any* answer for. `OOA` counts as an answer.

Accuracy Of the answers given, the share that are right.

Correct / wrong The same two quantities as counts, which is what makes settings comparable. Coverage and accuracy always move in opposite directions, and neither ratio alone says whether an exchange was worth taking.

Score `correct - k × wrong`, with `k = 2`. A judgement, stated so it can be disagreed with: a wrong aerodrome is worse than a silence because it corrupts *two* movement counts — the one it invents and the one it omits — and is invisible downstream, where a silence corrupts one and is an explicit null anyone can filter on.

A change is therefore worth taking exactly when it wins more than `k` correct answers per wrong one it introduces.

Out-of-area recall and precision are reported separately, because a method could otherwise buy headline accuracy by labelling `OOA` liberally. Recall is the share of genuinely out-of-area flights labelled as such; precision is the share of `OOA` labels that are correct. Where the report says *per-aerodrome recall* later on, that is a different measure and is defined where it appears.

The 2025 sample carries **95,116** reference flights and the 2024 sample **92,799**.

5 What each change is worth

Each row adds one change to the row above **and keeps it**, so the marginal worth of a change is the difference between adjacent rows, and the last row is the shipped pipeline exactly. That is asserted before the run rather than checked afterwards: the harness refuses a ladder whose final rung is not `DetectionConfig()` in its thresholds, its per-role algorithms and its track table, and refuses one containing a rung that changes nothing.

Read the *signs* across the two periods before the sizes. A change that helps on one sample and hurts on the other has not been demonstrated, whatever its magnitude on the sample it was tuned against.

5.1 2025

Step	Departures	Δ dep	Arrivals	Δ arr	Total
legacy algorithm, uncleaned tracks	63,064	—	60,600	—	123,664
+ rank candidates by exact distance	62,539	−525	60,255	−345	122,794
+ cut the radius by distance, not hexagon band	62,512	−27	60,255	0	122,767
+ smooth altitude before the flight-level cut	62,714	+202	60,211	−44	122,925
+ scheduled-service penalty	63,563	+849	61,072	+861	124,635
+ tuned flight-level cap	67,185	+3,622	61,748	+676	128,933
+ tuned vote margin	67,704	+519	62,078	+330	129,782
+ bearing tie-break	67,899	+195	63,272	+1,194	131,171
+ departures from `endpoint`	71,778	+3,879	62,183	−1,089	133,961
+ tuned endpoint radius	71,985	+207	62,183	0	134,168
+ cleaned tracks (= **shipped**)	71,983	−2	62,185	+2	134,168

Scores, not ratios: coverage and accuracy always move in opposite directions, and only the counts say whether an exchange was worth taking. Net over the whole ladder: **+10,504**.

The first eight rows produce both roles with `trend`, so each change is attributable to its own line. The ninth switches departures to `endpoint`, which is why the departure column jumps there and barely moves before it — every trend parameter above is tuned for arrivals.

5.2 2024

Step	Departures	Δ dep	Arrivals	Δ arr	Total
legacy algorithm, uncleaned tracks	53,569	—	48,411	—	101,980
+ rank candidates by exact distance	53,335	−234	48,183	−228	101,518
+ cut the radius by distance, not hexagon band	53,326	−9	48,178	−5	101,504
+ smooth altitude before the flight-level cut	53,591	+265	48,200	+22	101,791
+ scheduled-service penalty	53,765	+174	51,482	+3,282	105,247
+ tuned flight-level cap	57,115	+3,350	53,178	+1,696	110,293
+ tuned vote margin	57,587	+472	53,631	+453	111,218

Step	Departures	Δ dep	Arrivals	Δ arr	Total
+ bearing tie-break	58,163	+576	54,234	+603	112,397
+ departures from `endpoint`	62,982	+4,819	53,253	−981	116,235
+ tuned endpoint radius	63,048	+66	53,253	0	116,301
+ cleaned tracks (= **shipped**)	63,046	−2	52,904	−349	115,950

Scores, not ratios: coverage and accuracy always move in opposite directions, and only the counts say whether an exchange was worth taking. Net over the whole ladder: **+13,970**.

The first eight rows produce both roles with `trend`, so each change is attributable to its own line. The ninth switches departures to `endpoint`, which is why the departure column jumps there and barely moves before it — every trend parameter above is tuned for arrivals.

6 The parameter sweeps

Both periods run on identical grids, and cells are ranked by the two together — each period's score divided by its own reference count so the larger sample cannot dominate. A single-period argmax appears only as a stability check, because over hundreds of cells the top cell moves under noise even when the surface underneath is stable.

Departures — best cells on both periods together

Rank	Flight-level cap	Vote margin	Radius (NM)	2025 score	2024 score
1	FL125	0	30	70,249	59,562
2	FL125	0	20	70,295	59,488
3	FL100	0	20	69,978	59,643
4	FL125	0	40	70,026	59,365
5	FL125	2	20	69,995	59,374

Arrivals — best cells on both periods together

Rank	Flight-level cap	Vote margin	Radius (NM)	2025 score	2024 score
1	FL60	0	30	61,183	52,752
2	FL75	0	30	60,947	52,943
3	FL60	0	40	61,124	52,713
4	FL75	0	40	60,860	52,909
5	FL60	0	60	61,013	52,588

The endpoint rule, which is what departures use

Role	2025 best	2024 best	Agrees
Departures	30 NM / 15,000 ft	30 NM / 15,000 ft	**yes**
Arrivals	30 NM / 10,000 ft	30 NM / 10,000 ft	**yes**

6.1 The flight-level cap, followed to the top

Flight-level cap	2025 departures	2025 arrivals	2024 departures	2024 arrivals
FL25	58,898	60,242	50,331	49,052
FL40	65,086	62,068	55,464	52,071
FL50	66,699	62,893	57,141	53,216
FL60	67,531	63,301	57,963	53,910
FL75	68,533	63,579	58,745	54,624
FL100	69,272	62,330	59,328	53,612
FL125	69,461	60,410	59,367	52,311
FL150	69,088	58,632	58,772	50,678
FL175	68,253	56,528	58,020	48,395
FL200	67,008	54,186	56,484	45,967
FL225	65,359	52,135	54,881	43,558
FL250	63,178	49,900	53,022	40,865
FL275	61,286	48,022	51,289	38,582
FL300	58,922	45,014	48,951	35,687

At 30 NM — the shipped radius — through the pipeline itself. Departures peak at FL125 and arrivals at FL75 on the 2025 sample. Coverage keeps rising with the cap — a higher cap admits more fixes — but accuracy falls faster, because fixes taken higher up are further from whichever aerodrome turns out to be the answer.

7 What ships, and how well it works

Configuration	Departures by	Arrivals by	Dep. coverage	Dep. accuracy	Arr. coverage	Arr. accuracy	Total
SHIPPED — `endpoint` departures, `trend` arrivals	endpoint	trend	79.97%	98.21%	68.29%	98.58%	134,168
`endpoint` for both roles	endpoint	endpoint	80.03%	98.14%	70.15%	96.22%	131,032
`trend` for both roles	trend	trend	73.51%	98.86%	69.81%	98.45%	130,832
legacy — the algorithm published before 2026	trend	trend	67.46%	98.82%	66.99%	98.19%	122,158
nearest aerodrome, unconditionally	nearest	nearest	91.64%	87.68%	88.55%	79.29%	86,837

Each row is a complete configuration — which algorithm names departures, which names arrivals — not a step in a sequence. The rows are alternatives to choose between.

All five read the **cleaned** tracks, so this table compares algorithms on one input. That is why the `legacy` row here does not match the ladder's first rung in Section 5: this one is the legacy *algorithm* on current input, and that one is the legacy algorithm on the input it actually shipped with. The pair separates the algorithm from the data, and neither number alone does.

The shipped configuration scores **134,168 against legacy's 122,158**, a gain of +12,010.

7.1 Which changes hold on both samples

A change measured on one sample has been observed; a change measured on two has been *demonstrated*. The ladder runs identically on both periods, so every rung can be graded on whether its sign survives.

Step	2025 Δ total	2024 Δ total	Verdict
+ rank candidates by exact distance	−870	−462	**consistently costs**
+ cut the radius by distance, not hexagon band	−27	−14	no material effect
+ smooth altitude before the flight-level cut	+158	+287	**holds**
+ scheduled-service penalty	+1,710	+3,456	**holds**
+ tuned flight-level cap	+4,298	+5,046	**holds**
+ tuned vote margin	+849	+925	**holds**
+ bearing tie-break	+1,389	+1,179	**holds**
+ departures from `endpoint`	+2,790	+3,838	**holds**
+ tuned endpoint radius	+207	+66	**holds**
+ cleaned tracks (= **shipped**)	0	−351	costs on 2024 only

Δ is the change in **total** score, departures plus arrivals, against the rung above. A rung moving less than 48 flights on both samples is reported as having no material effect rather than having its sign read off noise.

Nothing changes sign between the two samples. Every step that moves anything moves it the same way on two independent periods — which is the property the second sample exists to test, and which an earlier version of this configuration did not have.

+ cleaned tracks (= shipped) moves one sample and not the other. That is weaker evidence than agreement and stronger than a contradiction: nothing points the wrong way, but the effect is not established on both.

Consistently costs is not the same as *wrong*, and the two rungs carrying that verdict are worth reading rather than skipping.

Exact-distance ranking costs on its own. It is not an improvement; it is an *enabler*. Two rungs later the flight-level cap is worth over four thousand on both samples, and under the hexagon-index selection it replaces, the same change lost ground. The cost here buys that, and quoting this rung alone tells the story backwards.

Cleaning costs too, and the report does not hide it. Masking implausible values removes candidate fixes, so coverage falls. What it buys is accuracy and a trustworthy input for every other thing built on tracks. Whether that exchange is worth it at $k = 2$ is a judgement, and the number is here so a reader can disagree with it — see Section 11.

Out-of-area labelling reads as *holds*, which corrects an easy misreading: it does cost arrival score, and it gains more on departures than it costs on arrivals, because flights beginning outside the observed area are now named correctly instead of being silently dropped. Section 10 gives both halves.

7.2 Does it hold on the second period?

Configuration	2025 total	2024 total
legacy — the algorithm published before 2026	122,158	100,591
SHIPPED — `endpoint` departures, `trend` arrivals	134,168	115,953

No parameter was chosen against one period and checked on the other: both were swept on identical grids and ranked together.

8 Three values this study changed

Version 6 shipped three constants that version 7 measured and found wanting. All three are changed in the code this report describes, not merely noted: the whole point of measuring a default is to act on the answer.

Each was tested the same way — applied to the **shipped configuration on its own**, on both samples. That is a different question from a rung in Section 5, which asks what a change was worth at one point in a sequence. This asks the question that decides: should it be on?

Was → is	2025	2024
radius 20 NM → **30 NM**	+181	+334
out-of-area on `trend` on → **off**	+470	+14
vote margin 2 → **0**	+395	+296

Each figure is the change in total score from adopting the new value, measured against the shipped configuration on that sample. Both columns positive means the change is supported on two independent periods.

The radius. Version 6 moved it to 20 NM on a single period. On two it gains on the sample it was chosen against and loses on the other — the only step in this study whose sign did not survive an independent sample — and the research sweep ranked over both periods puts 30 NM at an interior optimum. Reverted.

It turned out to be holding two other changes back. At 30 NM the bearing tie-break is worth more, because a wider radius admits more candidate aerodromes and so more near-ties for alignment to resolve; the switch of departures to `endpoint` gains more as well. A value can be wrong not only in itself but in what it suppresses, and that is invisible until it is changed rather than tabled.

Out-of-area on `trend`. Built, measured, and left off. The capability is real — `trend` could not previously distinguish "outside the observed area" from "could not tell" — but in the shipped configuration it reaches only arrivals, where the label is right about half the time against `endpoint`'s 89%. Every label replaces a *silence*, so at $k = 2$ a coin flip is a losing trade.

The asymmetry is geometric rather than a tuning accident. A track that *begins* at the edge of the observed area almost certainly entered from outside; one that *ends* near it may equally be a flight still bound somewhere inside whose reception was lost. Section 10 gives the figures.

The vote margin. Version 6 shipped 2 on research-sweep evidence and called it the weakest value in its recommendation. The pipeline now agrees with the sweep, on both samples, and the exchange rate settles it: dropping the margin gains far more correct arrivals than wrong ones, well above the study's threshold of two.

Cleaning is why it stopped earning its keep. The margin exists to stop noise flipping a climb-or-descent call, and step 02a now masks about a fifth of barometric altitudes before the vote is counted. It was defending against something already removed. **That interaction could not have been found before this version**, because cleaning did not previously reach the flight list at all — which is the strongest argument in the study for having wired it in.

9 What is still open

Two things, stated so that nobody has to infer them from silence.

Out-of-area for arrivals needs a direction test. The border test asks whether a track *ends* near the edge. It should ask whether the track is heading *out* — a fix near the boundary on an inbound heading is a reception loss, not a departure from the area. The course is already computed for the bearing tie-break, so the machinery exists; the rule and its measurement do not. Until then arrivals take their out-of-area labels from `endpoint`, at 89% precision.

A third period, held out. Every parameter here is chosen on two samples and validated on those same two. A period against which nothing was tuned would say whether the choices generalise or merely fit both Junes. It needs committed Network Manager reference data for whichever month is chosen, which is a decision about data availability rather than about method.

10 Out-of-area, and the three states

Role	Truly out-of-area	Labelled out-of-area	Recall	Precision	In-area accuracy
Departures	8.30%	0.74%	7.96%	89.20%	98.58%
Arrivals	8.37%	0.00%	0.00%	0.00%	98.61%

In-area accuracy is accuracy restricted to flights whose true aerodrome is a real one — detection skill with the free out-of-area wins removed, and the number to read when comparing naming rules.

11 What cleaning costs and buys

Cleaning is the last rung of the ladder rather than a separate experiment, so what it is worth is measured *given everything else already adopted* — which is the only form of the question that bears on a decision. The two rows differ in nothing but the track table.

2025

Role	Raw coverage	Clean coverage	Raw accuracy	Clean accuracy	Δ score
Departures	79.97%	79.97%	98.21%	98.21%	−2
Arrivals	69.07%	68.29%	98.22%	98.58%	+2

2024

Role	Raw coverage	Clean coverage	Raw accuracy	Clean accuracy	Δ score
Departures	72.69%	72.69%	97.82%	97.82%	−2
Arrivals	62.75%	62.11%	97.15%	97.26%	−349

Cleaning **masks implausible values to null and keeps the row**, and the detection path drops fixes with no barometric altitude — so it can only remove candidate samples. Coverage falling and accuracy rising is the expected shape; whether that exchange is worth taking at `k = 2` is what the score column says.

12 The 5 s sampler

The bin-based rule is adopted because it is the **correct** rule, not because it measures better. It does not depend on the OpenSky archive being a complete 1 Hz grid, which is a property of the archive rather than a guarantee; the fixed-phase rule it replaces survives only because that property currently holds, and degrades badly if it ever stops.

> **i** This figure is not re-measured here, and why
>
> Version 6 compared the two samplers across every parameter cell of two complete runs — 371 cells of the trend grid and 126 of the endpoint grid, identical grids over identically-treated input — and found the largest median coverage difference anywhere to be a few *thousandths* of a percentage point, against a decision bar of 0.5.
> This report does not repeat that measurement, and says so rather than restating the number as if it had. There is no fixed-phase arm left to compare against — the bin-based rule is now what the pipeline does — and these sweeps run on cleaned tracks over a different flight-level grid, so a comparison against version 6's arm would be confounded by the cleaning and the grid rather than measuring the sampler. Two arms on different data is exactly the error that produced version 6's worst figure, and manufacturing a comparison here would repeat it.

Where the sampler *should* matter is not here at all. Aerodrome detection reads an altitude trend over tens of samples, and one extra fix per five-second bin cannot change the sign of a trend. Milestones defined by a single instant on the ground — off-block, take-off, landing, in-block — are a different case, where a recovered bin is the difference between having a fix and not. That measurement belongs to the events study and has not been made.

One consequence is not reversible by configuration: **`track_id` values differ** between the two samplers, because rescued rows sit at track boundaries and the splitter breaks on gaps.

13 The bearing tie-break

This is the newest component in the algorithm and the best evidenced. It was proposed by the research sweep, declined once for want of a pipeline measurement, and is adopted here because `process_dai` confirms it on both samples.

Among candidates whose distances differ by less than a small band — that is, already equidistant to within the measurement — the tie is broken by *alignment*: which aerodrome the aircraft's course actually points at. Outside the band, distance decides alone.

The band is the whole mechanism. Alignment cannot *name* an aerodrome, because every field on the same radial behind the right one is equally well aligned; it can only separate two that distance has already declared equal.

14 Using the columns

Filter on the source columns, not on nulls. `ADES_SOURCE == 'aerodrome'` is the set of named arrivals; `'out_of_area'` is the set correctly identified as leaving the observed area; `'undetermined'` is the set to drop.

Expect under-count, not over-count. Abstention is the dominant failure mode, so an aerodrome's movement count from OPDI is a lower bound. Section 15 gives the ratio per aerodrome for the hundred busiest.

Join on `ICAO24` plus trimmed `FLT_ID` plus date. Callsigns are space-padded to eight characters in the raw feed.

`FIRST_SEEN` is not off-block. It is when reception began.

Do not compare `ID` across versions. See Section 16.

15 By aerodrome, for the busiest hundred

Being regenerated.

The matching is flight by flight, not count against count. **Per-aerodrome recall** is, of the flights whose *true* aerodrome is X, the share OPDI attributes to X. **Count ratio** is what OPDI assigns to X divided by X's true movements — it includes flights that truly belong elsewhere, so it is what an operational count would show, and it can exceed one.

Role	Movements in truth	Recall, shipped	Recall, legacy	Median count ratio
Departures	65,382	91.52%	82.05%	0.987
Arrivals	65,360	80.14%	79.11%	0.904

16 Versions, and what is reproducible

The `VERSION` column says which algorithm produced a row.

Version	What it means
`v2.0.0`	Legacy `trend`, both roles.
`v3.0.0`	Legacy `endpoint`. In a merged list these two were coalesced into one column, so a row's version told you about the departure side only.
`v4.0.0`	This study: exact-distance ranking, per-role algorithms, tuned thresholds, the bearing tie-break, out-of-area labelling, and cleaned tracks as input. One string for the whole flight list.

`DetectionConfig.legacy()` still returns every pre-2026 value, so re-processing a month reproduces the aerodromes a release published.

> ⚠ `track_id` is not reproducible across the sampler change
>
> The 5 s thinning rule changed at the same time. Rescued rows sit at track boundaries and the splitter breaks on gaps, so a track that gains a fix can be split differently. Re-ingesting an old month reproduces the *aerodromes* but not the *identifiers*. Do not join `ID` across releases.

17 Limitations

What is known to be missing or fragile, stated rather than implied.

- **Every shipped value is chosen on the same two samples it is validated on.** Nothing here is held out. Section 9 says why a third period would be worth having and what it needs.
- **The benchmark's own alignment is fragile where the flight list gains rows.** Each reference flight is paired with the track starting closest to its off-block time, so a configuration that produces *more* tracks can lose a pairing to one carrying no answer. This is visible wherever departures and arrivals come from different methods, and it is reported rather than corrected away: a benchmark that quietly adjusts its own alignment to flatter a result is worth less than one that shows where it is fragile.
- **The flight-level grid is swept at one radius.** The cap and the radius are measured separately rather than jointly in the pipeline — the research sweep covers the full grid, and the pipeline confirms the cap at the shipped radius only. A genuine interaction between the two would not be visible here, and one has already been found elsewhere in this study: the detection radius changed what the bearing tie-break was worth.
- **The cleaning comparison is one-sided.** Cleaning is measured as the ladder's last rung, and departures show no change there — `endpoint` reads a candidate cache built from cleaned tracks in every rung, so the earlier rungs were never really "raw" for that role. The figure is the cost to arrivals, not to the pipeline as a whole.
- **The legacy algorithm is not exactly reproducible.** Two identical runs of it named three arrivals differently out of 95,116. Ring-count selection leaves candidates tied on an integer, and the tie is resolved by an ordering that does not separate them. Exact-distance ranking removes those ties rather than arbitrating them, and every configuration using it reproduced exactly.
- **The candidate set is large and medium aerodromes.** A flight whose true aerodrome is a small field or a heliport cannot be named by any method here, whatever its thresholds, because the aerodrome is not a candidate. These flights are counted in every denominator on this page, so each coverage figure already carries their cost.
- **The border margin is fixed at 30 NM.** It decides what counts as leaving the observed area. Changing it redefines the ground truth as well as the method, so a sweep would move the target along with the aim.
- **H3 resolution 7 is fixed.** It no longer decides which candidate wins — that is exact distance now — but it still decides which aerodromes become candidates.
- **Two three-day samples.** Both in June. Seasonal effects are not measured.
- **Ground truth covers about ninety aerodromes.** Per-aerodrome figures are meaningful only for those.

18 Provenance

Every CSV behind this page carries the script, arguments, commit and source-file fingerprint that produced it, in `data/_manifest.json`. An output with no entry is reported as unverified rather than shown as fact.

18.1 Where reproducibility stops

Everything on this page can be recomputed from the OpenSky archive and the code in the `opdi` repository, with one exception: the EUROCONTROL Network Manager movement data used as ground truth is not redistributable. It is an external input, and every accuracy figure here is measured against it.